

Immunotherapy for lung cancer

Immunotherapy drugs are used to treat some non-small cell lung cancers (NSCLC) that have spread outside the lung or to other parts of the body. Less commonly, they may be used to treat SCLC.

On this page

- [About immunotherapy for lung cancer](#)
- [Tests on lung cancer cells](#)
- [How immunotherapy drugs are given](#)
- [When are immunotherapy drugs given?](#)
- [How long do I have immunotherapy for?](#)
- [Which immunotherapy drugs may be used?](#)
- [Side effects of immunotherapy for lung cancer](#)
- [Getting support](#)
- [How we can help](#)

About immunotherapy for lung cancer

Cancer drugs are often used to treat lung cancer. [Immunotherapy](#) drugs are a type of cancer drug. They make the immune system become more active. This makes the drugs better at finding and attacking cancer cells. The immune system protects the body against illness and infection.

These drugs are more commonly used to treat [non-small cell lung cancer \(NSCLC\)](#). Some immunotherapy drugs can be used to treat [small-cell lung cancer \(SCLC\)](#). You might have an immunotherapy drug with or after other treatments.

Some people may have an immunotherapy drug in a [clinical trial](#).

Booklets and resources



01 Jan 2025

Targeted and immunotherapy for lung cancer

Tests on lung cancer cells

You might have tests on the lung cancer cells to check which cancer drugs are more likely to work for you. This is called molecular testing or biomarker testing.

Tests can check whether the cancer cells have high levels of a protein that prevents the immune system from being able to attack the cancer cells.

If the lung cancer cells have enough of a protein called PDL1 the cancer is likely to respond to immunotherapy. Immunotherapy drugs can still be given even without a positive PDL1 result so not everyone needs a test.

There are different [types of immunotherapy drugs](#). Checkpoint inhibitors are the most common type used to treat [non-small cell lung cancer \(NSCLC\)](#). Checkpoint inhibitors are sometimes called CPIs for short.

They are usually used to treat advanced lung cancer and sometimes locally advanced lung cancer.

How immunotherapy drugs are given

A nurse usually gives you immunotherapy drugs through a drip into your vein. This is called an infusion. An immunotherapy drug called [atezolizumab](#) (Tecentriq[®]) can also be given as an injection under the skin (subcutaneous).

We have more information about [how cancer drug treatments are given](#).

When are immunotherapy drugs given?

Immunotherapy drugs can also be given:

- on their own
- together with chemotherapy drugs
- with a [targeted therapy drug](#) or another immunotherapy drug
- after lung cancer [surgery](#) or sometimes before surgery with [chemotherapy](#).

How long do I have immunotherapy for?

You might be able to have these drugs for up to 2 years. This depends on any of the following reasons:

- How helpful the drug is in controlling the cancer.
-

How the side effects affect you.

- Why you are having immunotherapy. For example, if you have immunotherapy after surgery to reduce the risk of the cancer coming back (adjuvant treatment) you usually have it for 1 year. If you are having it before surgery (neo-adjuvant treatment) you have it over several weeks.

If side effects are making you unwell, the drug might need to be reduced or stopped for a time until you feel better. If you have serious side effects your doctor may advise stopping the treatment. They can then talk about other possible treatment options for you.

It is normal to worry about a treatment coming to an end. But immunotherapy drugs can keep working for a time even after treatment is stopped. You will then be monitored closely with tests to make sure the cancer is not growing.

Which immunotherapy drugs may be used?

This depends on the type of lung cancer and stage of lung cancer that you have.

Non-small cell lung cancer

If you have non-small cell lung cancer you may have one of the following drugs:

Pembrolizumab (Keytruda[®])



[Pembrolizumab](#) may be your first treatment if the cancer cells show high levels of the PD-L1 protein. This is because it is more likely to work well for you.

If you have locally advanced lung cancer that can be removed with surgery, you might also have pembrolizumab. But this is not yet standard treatment.

Atezolizumab



[Atezolizumab](#) may be given:

- as a first treatment on its own, if you cannot have chemotherapy
- after treatment with chemotherapy
- with a targeted therapy drug called [bevacizumab \(Avastin[®]\)](#) and chemotherapy
- if you have already had an [EGFR or ALK targeted therapy drug](#), or if they are not suitable for you
- after surgery for some early and locally advanced cancers, to reduce the risk of lung cancer coming back (adjuvant treatment).

Durvalumab (Imfinzi[®])



[Durvalumab](#) might sometimes be given as ongoing treatment to control the cancer. This is sometimes called maintenance treatment. You might have durvalumab if you have had chemotherapy and radiotherapy given together and it has worked well.

You have durvalumab every 4 weeks for up to 1 year.>

Nivolumab (Opdivo[®])



[Nivolumab](#) can be given in the following ways:

- You may have it on its own after you have already had treatment with chemotherapy, usually every 2 or 4 weeks.
- You may have it before surgery, with chemotherapy to shrink the cancer. This is usually when the cancer has spread to the lymph nodes or there are bigger tumours in the lung. You will be given nivolumab usually every 3 weeks for up to 3 cycles of treatment.

Ipilimumab (YERVOY[®]) and nivolumab



These 2 immunotherapy drugs may be given together, but this is usually in a [clinical trial](#).

Small cell lung cancer

If you have [extensive stage small cell lung cancer](#) you might be given either of these immunotherapy drugs with chemotherapy as your first treatment:

- atezolizumab
- durvalumab (Imfinzi[®]).

This is only if you have not had any other treatment for the cancer. You also need to be well enough to have it.

Durvalumab is being assessed by the National Institute for Health and Care Excellence (NICE) as a possible treatment for [limited stage SCLC](#). NICE monitors which drugs and treatments are available on the NHS in England.

Other immunotherapy drugs, such as serplulimab, might also be used or be given in a clinical trial.

Side effects of immunotherapy for lung cancer

Different immunotherapy drugs can cause different side effects. Your cancer team will give you information about possible side effects before you start immunotherapy treatment. It is important to follow the advice they give you.

Immunotherapy drugs can cause immune-related sided effects as well as general side effects. This depends on the immunotherapy drug you are having.

Sometimes immune side effects can be very serious. Immunotherapy drugs can make the immune system react by attacking healthy tissue. This can result in inflammation in different organs of the body.

Although serious side effects are less common, it is very important to know about them. These types of side effects can start in the first weeks or months of treatment. But immune system side effects can happen sometimes up to 2 years after you finish treatment. They need to be treated quickly. If you think you have any of these side effects, contact the hospital straight away on the 24-hour number they give you.

Sometimes the drug might need to be stopped for a time until side effects improve. If you have serious side effects your doctor may advise stopping the immunotherapy treatment permanently. If this is needed, your doctor can talk to you about other possible treatment options for you.

We have more information about [immune system side effects and general side effects of immunotherapy drugs](#).

We also have more information about [individual immunotherapy drugs](#).

Getting support

Macmillan is also here to support you. If you would like to talk, you can:

- Call the Macmillan Support Line for free on [0808 808 00 00](tel:0808 808 00 00).
- [Chat to our specialists online](#)
- Visit our [lung cancer forum](#) to talk with people who have been affected by lung cancer, share your experience, and ask your questions.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our lung cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

National Institute for Health and Care Excellence (NICE). Lung cancer – Diagnosis and management. Clinical guideline 2019. Last updated 2023. (accessed Nov 2023) Available at: <https://www.nice.org.uk/guidance/ng122>

European Society for Medical Oncology (ESMO). Small-cell lung cancer: ESMO clinical practice guidelines for diagnosis, treatment and follow-up. 2021. (accessed Nov 2023). Available at: <https://www.esmo.org/guidelines/guidelines-by-topic/esmo-clinical-practice-guidelines-lung-and-chest-tumours/small-cell-lung-cancer>

European Society for Medical Oncology (ESMO). Early and locally advanced non-small-cell lung cancer (NSCLC): ESMO clinical practice guidelines for diagnosis, treatment and follow-up. 2017. eUpdate 01 September 2021: New Locally Advanced NSCLC Treatment Recommendations (accessed Nov 2023) Available at: <https://www.esmo.org/guidelines/esmo-clinical-practice-guideline-early-stage-and-locally-advanced-non-small-cell-lung-cancer>

European Society for Medical Oncology (ESMO). ESMO expert consensus statements on the management of EGFR mutant non-small-cell lung cancer. 2022 (accessed Nov 2023). Available at: <https://pubmed.ncbi.nlm.nih.gov/35176458/>

Cancer Information Development team

Author

Dr David Gilligan

Reviewer

Consultant Clinical Oncologist
Addenbrookes Hospital, Cambridge

Date reviewed

Reviewed: 31 January 2025 | Next review: 31 January 2028

Our cancer information meets the PIF TICK quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information.](#)

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
-

use short sentences

- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help